

The Nine Open Questions: Machine Closure

The OPEN9 (v19), v20, v21 and v22 campaigns: machines, machine facts, verdicts and honest caveats

Repository: github.com/wild8highlander/choptuik_ac_bc

Folder: `einstein_direct/open9`

Campaigns: OPEN9 v19, v20a, v20b, v21 (protocol executed), v22

Date: September 28, 2026

Abstract

This monograph records the machine closure of the nine open questions formulated in Chapter 7 of the verification monograph. For each question a dedicated deterministic machine was built in the repository style: hypothesis, machine confirmation or refutation, a JSON verdict, and honest caveats. The work is single-threaded (BLAS pinned to one core), avoids random number generation wherever possible, and treats reproducibility as a primary requirement: every number quoted here is reproduced by running the corresponding machine from the package.

The OPEN9 campaign (v19) produced nine machine verdicts: question No. 1 is closed by a proved structural barrier (the algebraic books cannot select $\pi/7$); No. 5 and No. 6 delivered two new exact machine facts (the identity $\tau(\delta_C) = \tau_{T1c}$ and the coincidence of the $45/8$ pole with the $\text{char}(2)$ root while the linear level stays finite); No. 7 is closed at accessible depth (no growing $1/r$ mode — a junk floor); No. 9 received Krawczyk certificates (zero new branches confirmed machine-rigorously); No. 4 is quantitatively blocked (background, not outlier); No. 2 and No. 3 were reduced to falsifiable measurement protocols. The v20 campaigns turned both protocols into executable specifications: the amplitude ladder was executed on the PDE machine with an honestly negative verdict (the exponent is not measurable on the current machine), and the depth $z \geq 30$ target received a closed-form linear-sector certificate, a floor-suppression budget, and certified estimators.

The full regression suite of the package is 58 pytest tests (21 core, 22 for the OPEN9 machines, 4 for the v20 campaigns, 5 for v21, 6 for v22). Every claim in this monograph carries explicit caveats about its domain of validity; no verdict goes beyond what the machines actually measured or proved.

The Section 5 protocol is executed by the v21 campaign: the gates, rollback, window tightening, checkpoints and both stop branches all fired or were exhausted legally; the headline fact — the depth wall is window-policy sensitive (z : 1.78 to 7.20 at $w = 3$) and the next wall is the v-budget. The v22 campaign exposed the topology of that wall (a coverage ratchet, dead v-extension, a vacuum final stage) and broke it with a root window anchor: the depth record $z = 8.56$ and the fourth echo peak are reached; the quartet's period is not DSS (3.44 excluded at 6.3 sigma) — periodicity at this depth is honestly unconfirmed.

Table of Contents

Note: this Table of Contents is generated via field codes. After editing the document, right-click the TOC and choose "Update Field" to refresh the page numbers.

1. Introduction

The choptuik_ac_bc repository develops a machine-verification programme: every substantive claim must be reproduced by a machine — a program with deterministic behaviour that saves its result as JSON and is covered by regression tests. The campaign ledger v6-v18 travelled from the first zoom PDE machine to symbolic derivations, nonlinear DAE marches, and certified censuses. This monograph describes the OPEN9 (v19) and v20 campaigns — a systematic attack on the nine open questions of Chapter 7 — together with the v21 execution of that protocol and the v22 break of the depth wall.

The style of every machine is the same. First a hypothesis is formulated with a falsifiable prediction; then a machine is built that measures or proves the prediction; the result is saved as JSON with a verdict and honest caveats; the key facts are pinned by tests. Single-threadedness is a principle: multithreaded LAPACK reshuffles the rounding order and destroys the reproducibility of bisection and horizon nucleation. The offline mode is also a principle: when saved data suffice, the machine does not re-run heavy PDE evolution but works with its traces.

The campaign architecture followed the dependency graph of the questions. The hub of the symbolic line is question No. 5 (the truncated tower): its infrastructure serves No. 1, No. 2, No. 3 and No. 8. Question No. 7 gates No. 3: until the $1/r$ mode is measured, the percent-level gamma cannot be separated from its contribution. Questions No. 4, No. 6 and No. 9 admitted independent attacks; No. 9 is a certification instrument that strengthens all the others. This honest routing of effort across the graph determined the final verdicts.

2. The Nine Open Questions: Statements and Machines

Below we state each question, describe the machine, list the key machine facts and give the status after v19-v20. The question formulations are canonical per Chapter 7 of the verification monograph; all numbers are taken from the JSON verdicts of the package machines.

2.1. Question No. 1: what selects $\delta_C = \pi/7$

The seven-fold echo structure and the Berry form of the gauge point to the selector $\delta_C = \pi/7$, yet none of the algebraic books explains this choice. The machine brick_selection.py ran a three-level null test. Level L1: the books $\kappa(\delta) = 2 - \delta^2/2$ and $\tau(\delta) = 27/(4 - \delta^2)$ leave δ FREE — the elimination ideal in $\mathbb{Q}[\delta]$ is trivial (the Groebner basis is one), i.e. the algebraic relations of the books impose no condition on δ at all. Level L2: a PSLQ search over $[1, d, \dots, d^8]$ with $d = \pi/7$ at 50, 100

and 200 digits finds no relation, while the control procedure recovers the known within-sector relations — the method works, the relation is absent. Level L3 is a theorem: π is transcendental (Lindemann-Weierstrass), $\pi/7$ is a nonzero rational multiple, hence transcendental; a root of any nonzero polynomial with rational coefficients is algebraic, therefore no algebraic book can have $\pi/7$ as a root.

Verdict: the algebraic tower of books alone cannot select $\pi/7$ — the structural barrier is proved, not scanned. Selection can only enter through the transcendental objects of the inventory (π in the $7\pi/30$ spinor quanta and the Berry form; $\ln(16/9)$ in the hexcycle clock). The remaining substantive candidates are finite-amplitude monodromy and geometric holonomy (the Berry connection); both require the PDE level.

Caveats: the barrier is established for algebraic books with rational coefficients; the Berry-screening observation (closure with a +0.062% residual) remains an observation, not a principle; the tick obstruction does not select (the ladder minimum sits at $k = 12$, non-monotonically).

2.2. Question No. 2: the phantom residue to $\pi/30$

The phantom frequency $\text{Im } \lambda = 0.107731$ exceeds $\pi/30 = 0.104720$ by +2.88% (the pencil analogue by +3.10%) while the spread between two independent systems is 0.22%: the shift is structural, roughly thirteen times the systematic. The machine `finite_amplitude_mono.py` proved the residue incompatible with the linear sector: the residue $3.0\text{e-}3$ is $1.1\text{e}11$ times the march growth bound per echo (and $1.1\text{e}5$ times the $\sqrt{\epsilon}$ defectiveness floor) — linear monodromy cannot carry such a shift, in agreement with the exact annihilation in the linear world ($\Delta_{\text{mono}} = 0$, growth per echo below $2.8\text{e-}14$). The carrier candidate is the defective Jordan-2 block ($B^2 = 0$ exactly): the minimal model gives $\lambda(\epsilon) = \pm\sqrt{\epsilon n_{12} n_{21}}$, i.e. the frequency scales as $\epsilon^{1/2}$; a semisimple pair (a Lyapunov center) predicts an A^2 shift.

Verdict: the residue is a finite-amplitude phenomenon with a falsifiable discriminator: $p = 1/2$ (Jordan-2) versus $p = 1$ (the linear b2 exit, v15: $|Dr V| = (39.2 \pm 0.2)A$) versus $p = 2$ (a Lyapunov center). The exponent requires an amplitude ladder — see Section 3.1 for its execution: on the current PDE machine the exponent is not measurable, and the discriminator survives as a protocol.

Caveats: the $\sqrt{\epsilon}$ law is a model-level conclusion (the 2×2 minimal model is illustrative; the coefficient does not transfer to the full system), not a PDE-level one; both $\text{Im } \lambda$ values come from a `sigma_min` filter without guarantees; the consistency with question No. 1 survives — a finite-amplitude correction to a transcendental constant does not contradict the barrier.

2.3. Question No. 3: percent-level gamma/Delta

The literature values $\gamma = 0.374$ and $\Delta = 3.44$ do not separate from systematics on the repository data: OLS over twelve supercritical points gives $\gamma = -0.031$, the lower mass half of the sample is flat, and the medians of the lower and upper halves differ by a factor 0.79 — the mass floor dominates the whole sample. The machine `echo_fit_global.py` delivered and self-tested percentile estimators: on synthetic data with 35% of the lower masses clamped, OLS drags γ down to 0.281 against the true 0.374, while the percentile fit recovers 0.398 with a confidence interval [0.375, 0.420]. For Δ , the literature value and the theoretical $\Delta_{\text{cyc}} = 6 \cdot \ln(16/9) = 3.4522$ differ by 0.35% — the percent level is already consistent, but re-measuring it from echo trains is blocked by the depth wall.

Verdict: the method is delivered and certified; the numerical gamma on the saved data remains floor-limited. The question closes to the level of an executable specification: the certification of the linear sector to $z = 30$, the floor-suppression budget $S_{\text{req}}(30) \sim 2.26e3$ and the closed budget-estimator loop are delivered by v20b (Section 3.2); the full numerical verdict belongs to campaign v21.

Caveats: the bisection sample is small and clustered near A^* ; the floor $M \sim 4 \cdot du$ is not measured directly (the diagnosis is via the flat lower half); the masses are horizon-frozen, their relation to the final black-hole mass is indirect; the No. 7 gate is one-sided — the absence of $M1$ growth does not by itself guarantee percent-level accuracy.

2.4. Question No. 4: the W2 ring to 4/3

Symbolically the ring $W2/t0^2$ to $4/3$ is verified by three routes, but on the PDE data the channel is swamped by junk. The machine `mirror_ring_v2.py` re-analysed offline all 10,665 saved mirror pairs over 172 rows with a robust MAD triage: trimming outliers does not expose a signal core — the trimmed medians of $W2/t0^2$ are 1.15, 0.071 and 0.165 against the predicted $4/3$; the junk is not outlier junk, it is background. The core criterion (at least five rows with $|W2/t0^2 - 4/3| \leq 0.1$) fires on 6 of 172 rows, while the bootstrap null gives 5.3: the observation is indistinguishable from chance.

Verdict: the channel is QUANTITATIVELY BLOCKED at $z \leq 9.35$ — not for lack of data but because of background odd junk; repeating the analysis on the deep v21 chain with floor suppression is the only route to a measurable ring.

Caveats: the triage is robust to the metric choice (MAD and the interquartile range were both tried); the odd/even separation of 0.28 shows separability but not signal level; the conclusion does not extend beyond the wall.

2.5. Question No. 5: truncated-tower convergence

The machine `sympy_center_o7.py` established: the chain closes the linear z -system exactly — $F1/F2/F4/F5$ vanish identically in $T0h$, and $F3$ gives the linear closure $P4h_lin = t/10 + 16t^3/135$, a polynomial without singularities to which the $45/8$ pole does not belong. The $M3$ link $O4$ on the chain equals $2t^{2/3}$ — exactly the $O3$ route at $\kappa = 2$: the weight structure is consistent across levels. The naive $O6$ is reproduced as machine-incompatible ($\tau^3 = 9/4$ versus $1/\tau^5 = 9/16$ — an overdetermined system). The new exact fact is the identity $\tau \cdot (\delta_C = \pi/7) = 27/(4 - \pi^2/49) = 1323/(196 - \pi^2)$: the pair clock on the surviving brick and the one-brick $T1c$ closure coincide exactly — two independent routes, one number.

Verdict: the tower convergence to the exact clocks is not established (levels 6.75 to 7.1079, a +5.27% shift); the question is partially closed — the $O7$ programme with five fixations is written, and the new identity ties two independent book sectors together.

Caveats: the identity is established on the baseline and one brick; its transfer to an arbitrary brick is open; the level convergence is an open programme, not a theorem.

2.6. Question No. 6: the soft pole at $45/8$

The machine `puiseux_soft_pole.py` wrote out the chain $P4h_chain(t) = t^3(352t^2 - 540)/(135(8t^2 - 45))$: a simple pole in τ at $45/8$ with a nonzero numerator $6075 \cdot \sqrt{10}$ and an explicitly computed principal part of the Laurent series. An exact coincidence was found: the characteristic polynomial $\text{char}(\lambda = 2, \tau) = 8(2\tau - 1)(8\tau - 45)/27$ — the growing mode $\lambda^+ = 2$ appears exactly at $\tau = 1/2$ and $\tau = 45/8$, and the second point coincides with the chain pole: the mode and the pole are one structure. Softness is shown symbolically: the linear level at the pole is finite ($13 \cdot \sqrt{10}/40$); the divergence lives entirely in the nonlinear addition — the pole is an artifact of the chain representation, not a physical divergence.

Verdict: the $45/8$ pole is closed — a simple representation pole coinciding with the birth point of the growing mode; physical quantities at the pole are finite. The brick shift ($45/8$ to 5.9233 for one-brick) is recorded; transferring the $\text{char}(2)$ relation to an arbitrary brick is open.

Caveats: the pole-mode relation is proved on the baseline; the relation to the amplitude-linear $b2$ exit ($v15$) is a separate object — the mechanisms must not be conflated.

2.7. Question No. 7: the $1/r$ mode

The machine `m1_tracker.py` ran the decomposition $m = M1 \cdot \xi + M3 \cdot \xi^3$ over twenty saved profiles: the two-parameter form is stable on clean rows (median $R^2 = 0.999$;

18 clean, 2 flagged). The M1 regime is a junk floor: $|M1|$ decays monotonically with zoom depth (1.2e-8 to 1.1e-9 to 2.3e-12 to 4.8e-20; the log-linear slope is -8.35 with interval [-9.88, -6.47]). The junk invasion is localised: the fit degrades only in rows $z3/j12$ and $z3/j13$ — dirty stages are dropped but reported per the honesty protocol.

Verdict: at accessible depth $z \leq 9.35$ there is NO growing $1/r$ mode — $|M1|$ decays with zoom; the gate word for question No. 3 is issued: open-for-percentile (the gamma is floor-limited, not mode-limited). Full closure beyond the wall belongs to the depth ladder v21.

Caveats: the verdict is bounded by the accessible depth; the floor itself does not guarantee gamma cleanliness; beyond the wall the mode behaviour was never measured.

2.8. Question No. 8: the finite-amplitude DSS cycle

The machine `lyapunov_center.py` checked the local route to a cycle: the characteristic polynomial $\text{char}(i*\omega, \tau^* = 27/4)$ has the single real solution $\omega = 0$ — there are no purely imaginary pairs, and the corrected pencil agrees; the Lyapunov center theorem is inapplicable. The flat line is a line of exact equilibria; the b2 exit is amplitude-linear (39.2 ± 0.2)*A, so there is no local cycle bifurcation on the line either. The return ruler over the book-walk (2×6 stations) shows collapse — a monotone relaxation of $|R1h - 1|$ to zero with a return defect of the cycle up to 0.309 — a transient, not an attractor.

Verdict: the finite-amplitude DSS cycle, should it exist, is a GLOBAL object outside the reach of local analysis; the residue monodromy is computable only from the Poincare map of the PDE machine (the v21 protocol). The local route is closed machine-wise.

Caveats: the analysis is local at x^* ; global components of the solution set far from the critical point were not searched (shooting/nsolve — the next campaign); the stiffness is a property of the truncated tower.

2.9. Question No. 9: off-chain statics

The machine `krawczyk_census.py` built its own interval arithmetic with outward rounding and the Krawczyk operator (incidentally fixing a defect of the classical scheme: the residual must be evaluated at the box midpoint, not on the box). The one-dimensional clock census yields exactly one certified root box per model — [2.59806, 2.61861] for the baseline with the booked $T0h = 3*\sqrt{3}/2 = 2.5980762$ inside, and [2.66607, 2.66779] for one-brick with 2.6660683 inside — with uniqueness certificates and OUT-certificates everywhere outside the boxes. The two-dimensional census of the chain slice $G = (M_{\text{def}}, UV)$ over $[0.05, 10] \times [0.02, 10]$ gives an OUT coverage of 100.00% of the area; the undecided clusters sit exactly on the known non-isolated families $\{R1h = 0\}$ and $\{t = 3*\sqrt{3}/2\}$.

Verdict: the census is CERTIFIED — the zero new branches of v18b are confirmed machine-rigorously within the box and the slice; question No. 9 is closed in certified form.

Caveats: the census covers the pure clock and the chain slice, not the full eight-amplitude book; extending it to the full cube is a direct continuation of the same machine.

3. The v20 Campaigns: from Protocols to Measurements

Two protocol-level questions (No. 2 and No. 3) demanded a measurement level: an amplitude ladder for the exponent p and a depth $z \geq 30$ for the percent-level gamma. The v20a and v20b campaigns executed both protocols as far as the current machine allows and reduced the rest to executable specifications.

3.1. v20a: the amplitude ladder

The machine `amplitude_ladder.py` executed a real ladder over the zoom PDE machine: ϵ s in $\{3e-3, 1e-3, 3e-4, 1e-4\}$ above A^* (grid $n = 600$, up to eight zooms per rung, 15-20 seconds per rung). At each rung the curvature peaks $Q(v)$, the echo period Δ_{eff} via a geometric fit of the intervals, and the wiggle harmonic with a free frequency were extracted. The result: echo trains do not develop at any rung (0-3 significant peaks; stop criteria singularity and $v_{\text{exhausted}}$; depth $z \leq 4.6$). A cross-check against all saved campaigns (v2, v3.2, v5, the spinor ladder) shows the same: every run at the same amplitudes stops at singularity with $z \leq 6.05$ and zero peaks.

Verdict v20a: the exponent p is NOT measurable on the current machine — the absence of an echo train is a property of the machine (the depth wall), not of the chosen ϵ s. The discriminator survives as a falsifiable protocol ($p = 1/2, p = 1, p = 2$); the per-rung cost is negligible; the blocker is the wall physics, i.e. the v20b [D2] budget — first depth (v21), then the ladder.

Caveats: the rungs ran at $n = 600$ with a zoom cap; a deeper per-rung grid may shift the picture marginally; the relation of Δ_{eff} (tracking units) to the spinor $\Delta = 7\pi/30$ is mediated by the clock conversion — the discriminator is built on relative shifts and is robust to calibration.

3.2. v20b: depth $z \geq 30$

The machine `depth_z30.py` delivered four machine facts. [D1] The linear sector is certified to $z = 30$ in closed form: the M2 propagator is exactly $\Pi(T) = I + B^*T$ ($B^2 = 0$ — the Jordan-2 of the v14 closure), the drift phase is exactly zero (real spectrum $\{1, 1\}$), and the growth is secular: $\sigma_{\text{max}} \sim 126$ at $z = 30$ and ~ 4.4 per echo — polynomial, not

exponential; an RK4 march to $z = 30$ matches the closed form to machine precision. Consequence: a percent-level γ measured at $z \geq 30$ is clean of linear-monodromy contamination by construction — the linear sector carries neither a $\pi/30$ -ladder oscillation nor an exponent.

[D2] The depth budget: the mass floor must be suppressed by a factor $S_{\text{req}}(z) = \exp(\gamma(z - z_{\text{wall}}))$; at $z = 30$ this is $\sim 2.26e3$ (with a tenfold percentile margin $\sim 2.3e4$). The depth spans 41 spinor echoes. Bisection cannot carry depth: $(A - A^*)/A^* \sim \exp(-z/\gamma) \sim 1e-35$ at $z = 30$ — beyond double precision; the zoom rescaling does carry it ($z \pm \ln \lambda$ per stage); a realistic start is $\epsilon \sim 1e-3$ ($z_0 \sim 2.6$) plus 25-30 zooms.

[D3] The closed loop on the reduced ladder: synthetic echo trains (geometric peak intervals, masses $\exp(-\gamma z)$, the floor at depth $z_f = 7 + (z - 9.35)$, 5% junk on clamped points, fixed seeds) with an a-priori estimator rule (observed floor pile-up below 15% — survivor-OLS; otherwise percentile p70 of survivors) deliver a percent-level γ and a 0.5%-level Δ at all depths including $z = 30$, whereas the naive OLS reproduces the q3 failure: -23% at the wall and -2.5...-8% deeper. The budget-percent loop is closed: suppress the floor per [D2] and the estimators deliver.

[D4] The v21 protocol (Section 5) is recorded in the JSON verdict: start, junk gate, checkpoints, stop criteria. Caveats: the reduced model certifies the estimators and the budget, not the junk physics of the real machine; the floor-depth calibration $z_f = 7.0$ is anchored to the q3 floor-dominance diagnosis, not measured directly.

Verdict v20b: depth $z \geq 30$ is now a QUANTIFIED engineering target: a certified clean linear sector, a floor-suppression number ($2.3e3$ with margin), a certified a-priori estimator rule, and a written v21 protocol.

4. Status Summary and the Dependency Graph

The status summary after v19-v20 is given below. Each row is a machine verdict with its machine; the word closed always means a specific closure mechanism: a proved barrier, an exact identity, a certificate, a quantified block, or an executable protocol.

Table 1. Status of the nine questions after v19-v20

No.	Question	Machine	Status	Next step
1	selector $\delta_C = \pi/7$	brick_selection.py	closed: structural barrier proved	PDE level: monodromy / Berry holonomy
2	$\pi/30$ residue	finite_amplitude_mono.py	protocol + discriminator p	v21: ladder on the deep chain

No.	Question	Machine	Status	Next step
3	percent gamma/Delta	echo_fit_global.py + depth_z30.py	estimators certified; budget quantified	v21: $z \geq 30$ per [D2]
4	W2 ring to 4/3	mirror_ring_v2.py	quantitatively blocked (background)	v21: repeat on the deep chain
5	tower convergence	sympy_center_o7. py	partial: identity $\tau * (\delta_C) =$ $\tau * T1c$	the O7 programme (5 fixations)
6	pole 45/8	puiseux_soft_pole. py	closed: representation artifact	brick sensitivity of char(2)
7	1/r mode	m1_tracker.py	closed at accessible depth	v21: control beyond the wall
8	DSS cycle	lyapunov_center.p y	locally closed (no cycle)	v21: Poincare map
9	off-chain statics	krawczyk_census. py	closed: Krawczyk certificates	the full amplitude cube

The dependency graph was confirmed by the campaign itself: the No. 5 infrastructure served the symbolism of No. 1 and No. 6; the No. 7 gate word legally opened the percentile line of No. 3; the certification apparatus of No. 9 strengthened the census of static books; and the blocks of No. 3 and No. 4 reduced to the same physical cause — the depth wall and the mass floor — which fixed the v21 target.

5. The v21 Protocol: the Deep-Echo Campaign

The v21 protocol is the single missing step toward the numerical verdicts on No. 2, No. 3 and the ring of No. 4. Its schedule is recorded in the JSON verdict of depth_z30.py and consists of the following items.

1. Start: $\text{eps} = 1e-3$ above A^* ($z_0 \sim 2.6$), grid $n = 800$, zooms until the stop criterion.
2. Every stage: junk monitoring per the q7 gate word — while $|M1|$ decays with zoom depth, the open-for-percentile gate stays open.
3. Floor suppression: the W2/M3 gates active; on junk invasion ($j \geq 12$) — stage rollback and window tightening ($w_{\text{factor}} \times 5$).
4. Target pace: floor suppression via the gates, not via depth; a JSON checkpoint after every echo.
5. Stop: $z \geq 30$ (41 spinor echoes) or three consecutive junk invasions — in the latter case a new wall is recorded with a new S_{req} .

The v21 deliverables: echo trains to $z = 30$, a percentile gamma with a confidence interval, the depth evolution of Δ_{eff} and — with luck — an open $\pi/15$ modulation (one and a half periods). The amplitude ladder runs on the deep chain as a second pass: the per-rung cost is small; only the wall blocks.

5.1. Status: the protocol executed (campaign v21)

The v21 campaign (`deep_echo_v21.py`) executed the protocol end to end: the [P1]-[P5] branches all fired or were exhausted legally. Run 0 calibrated the gate: restart rows are vacuum ($m \sim 1e-13$, $R^2 < 0$) — a fit there is a record, not a diagnostic; the gate is active only on active rows ($Q > 1\%$ of the stage max). The main chain at the protocol defaults gave $z = 1.78$ (singularity; the wrapper is neutral to the physics — the control is exact). The [D-v21] window-policy probe exposed the headline fact of the campaign: the depth wall is trigger-policy sensitive — $z(w = 5) = 1.78$, $z(w = 3) = 7.20$ (deeper than every saved campaign, v20a included: 4.6), $z(w = 1.5) = 6.66$ — with zero junk invasions; the next wall is the v-budget (v_exhausted), not junk. The v20b [D2] budget is recomputed from the reached wall: $S_{\text{req}}(30) \sim 5.0e3$ ($\times 10$ margin $5.0e4$). The amplitude ladder as the second pass on the best policy is executed ($3e-3/1e-3/1e-4$), but the echo train does not develop (≤ 3 peaks per chain): the exponent p is honestly not measurable, and the $(1/2 / 1 / 2)$ discriminator is preserved as a falsifiable protocol.

Table 3. The v21 protocol execution (JSON: `v21_deep_echo.json`)

Protocol item	Execution
[P1] start	$\text{eps} = 1e-3$ above A^* , $n = 800$; control: the plain ZoomRunner reproduces the chain with zero deviation (the wrapper is neutral)
[P2] q7 junk gate	active on active rows; calibrated by run 0 (restart rows are vacuum)
[P3] rollback + tightening	the rollback machinery is ready and smoke-tested; zero invasions — the window lever moved into the [D-v21] probe
[P4] checkpoints	<code>v21_checkpoints.json</code> rewritten after every zoom/event
[P5] stop	the machine stop $z = 7.20$ (v_exhausted) is recorded as the new wall; $S_{\text{req}}(30) \sim 5.0e3$

5.2. Continuation: campaign v22 — the wall diagnosed and broken

The v22 order — extend the v-budget at $w = 3$ ($v_{\text{ahead_factor}} / \text{telescope}$) for the fourth peak and Δ_{eff} — is executed with a machine-directed redirect of the lever. The diagnosis (the v21 $w = 3$ chain reproduced with zero deviation): the telescope cap never binds (4/4 zooms free); the final stage is vacuum (the trigger ladder descends in v while the

window looks ahead into emptiness); v-extension is dead code by construction (slack exactly 0.0); the window floor can only rise (a coverage ratchet). The probes: v_ahead forward is inert; the global backward anchor is destructive (the restart row lands in the incoming-pulse tail $v_p = 0.5 \pm 0.1$); the late anchor is pinned by the floor. The fix — a root anchor (k1 on zoom 1 lowers the root floor below the expected echo ~ 0.4003) together with a coverage-floor anchor (from zoom 2): the depth record $z = 8.56$ (+19% over the 7.20 wall) and the fourth echo peak are reached (seven peaks, four on one stage). The quartet's $\Delta_{\text{eff}} = 0.611 \pm 0.446$ — the period is not DSS (3.44 excluded at 6.3 sigma): periodicity at this depth is honestly unconfirmed; the new wall is the root floor. Caveat: the v22 probes run the raw ZoomRunner physics without the [P2]/[P3] junk-gate wrapper; a full protocol run on the adopted policy is the next campaign.

6. Reproducibility and Package Layout

The open9 package is self-contained: the machines carry a private copy of the results and run from a single folder without repository paths. The standalone smoke suite is ten tests; the full repository regression is 58 pytest tests. All machines are single-threaded and deterministic; the reduced models of v20b use fixed seeds.

Table 2. The open9 package layout

Component	Contents
machines/	13 machines (9 OPEN9 + 2 v20 + v21 + v22) with transitive dependencies
machines/results/	a private copy of all campaign JSON verdicts (incl. v20a/v20b, v21, v22) and .npz bases
tests/	the standalone smoke suite (10 tests) and the machines with private results
monographs/	this monograph: RU/EN x docx/pdf (4 files)

Running: from the open9 folder — `python3 -m pytest tests/test_open9_standalone.py -q`; from the machines folder — any machine, e.g. `python3 depth_z30.py` (about a minute) or `python3 amplitude_ladder.py` (one to two minutes). The claims of this monograph are reproducible in full: every number is read from the corresponding JSON verdict, and every machine fact is pinned by a test.